MILITARY SPECIFICATION

INDICATOR, ATTITUDE, TYPE ARU-51/A

This amendment forms a part of MIL-I-87928(USAF), dated 1 NOV 80, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete "2.1" and add:
"2.1 Government documents."

* Add:
"2.1.1 Specifications, standards, and handbooks. The following specifications,standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Under SPECIFICATIONS, Federal, delete:
"QQ-P-416 Plating, Cadmium (Electrodeposited)".

* Under SPECIFICATIONS, Military, add: "MIL-M-7793 Meter, Time Totalizing".

* Change address in beneficial comments block to "Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990."

PAGE 2

* Under SPECIFICATIONS, Military, delete "MIL-B-83292", and add:
"MIL-L-25467 Lighting, Intergral, Red, Aircraft Instrument, General Specification For", "MIL-B-81793 Bearing, Ball, Precision For Instruments & Rotating Components" and "MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited".

* Under STANDARDS, Military, delete "MIL-STD-100", "MS17322" and add:
"DOD-STD-100".

PAGE 3

* Delete "2.2 and add:
"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the nongovernment documents which is current on the date of the solicitation.

A FM 03951 Packaging, Commercial


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(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Delete and add: "3.2 DATA. If required by the contracting activity the data and drawings shall be submitted in MIL-STD-831 and DOD-STD-100 format respectively."

PAGE 4

* Add:
"3.4.1.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

PAGE 6

Add 3.5.3 thru 3.5.3.2:
"3.5.3 Test equipment. Indicators furnished under this specification shall be capable of being completely functional tested with the test equipment listed below in paragraphs 3.5.3.1 and 3.5.3.2. Additional electrical interface boxes and/or interconnecting cables shall not be required."

3.5.3.1 Depot testing. Complete functional performance verification at depot level shall be accomplishable on tester, LSI P/N LT4735-04-01 and interconnecting cable. The technical order covering the tester is 33D3-17-29-1,4. The adapter and interconnecting cable are covered in technical order 33D3-25-44-1. Use of this tester adapter and interconnecting cable are described in technical order 5F8-3-38-3.

3.5.3.2 Field testing. Complete functional performance verfication at field level Shall be accomplishable on universal tester, Sperry P/N 1510340 (FSN4920-971-0204), plus tester adapter, LSI P/N LT5973-01-01 and interconnecting cable. It shall be possible to test all indicator functions, except the failure monitor functions, when testing the indicator on universal tester P/N 1510340 without tester adapter P/N LT5973-01-01 and interconnecting cable. The technical order covering universal tester P/N 1510340 is 33D3-17-9-1,4. Use of this tester, without the failure monitoring functions, is described in technical order 5F8-3-36-2."

PAGE 7

Delete "3.6.e".

PAGE 8

* 3.7 Line 4 delete "MIL-STD-100" and add: "DOD-STD-100".

PAGE 14

Table 1, under Function for Connector pin h delete 'spare" and add "Internal lighting power input for red lighting (0-5V) .

PAGE 17



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Table II, delete and add:

TABLE II. Amplifier connections

| (Connector shall mate with Bendix clamp PT01E-16-26P (SR), or equal, or potting PT1P-16-26P or equal) | | |
|---|---|---|
| Connector Pins | Functions | |
| A | 115 VAC, 400Hz | (Power) |
| B | Azimuth Trim | (Test Point) |
| C | Roll Generator | (Signal) |
| D | 26 VAC Motor-Generator Fixed 0 Winding | (Power) |
| E | Pitch Generator | (Signal) |
| F | Roll Motor | (Control 0) |
| G | Spare | |
| H | 18 VAC | (Power) |
| J | Pitch Trim | (Signal) |
| K | Azimuth Synchro | (Signal) |
| L | Azimuth Generator | (Signal) |
| M | AC Neutral - DC Common | |
| N | Pitch Common | |
| P | Spare | |
| R | Pitch Motor | (Control 0) |
| S | Azimuth Motor | (Control 0) |
| T | Roll Synchro | (Signal) |
| U | Azimuth Common | |
| V | Pitch Cosine | (Synchro Monitor) |
| W | Roll Common | |
| X | Pitch Trim | (Center Tap) |
| Y | Flag Power | |
| Z | Pitch Synchro | (Signal) |
| a | Roll Trim | (Test Point) |
| b | Roll Cosine | (Synchro Monitor) |
| c | Spare | |

PAGE 24

* 3.10.11 Line 2 delete "MS17322" and add: "MIL-M-7793".

PAGE 25

3.13 Line 6: delete "8" and add "8.6".

3.13.2 Line 2: delete "1.0" and add "5.0".

Delete 3.14 thru 3.14.2 and add:
"3.14 Lighting. Internal lighting shall be designed into the indicator. The design of the indicator shall be such that the internal lighting may be either white or red depending upon the pins (specified in table I) to which the power for lighting is applied. With the exceptions specified herein, the red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accoding with MIL-L-27160, except that the miniature airplane brightness shall range between 1.0 and 2.0 foot-lamberts.



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The ratio of the brightness of the gray portion of the sphere to the brightness of the miniature airplane shall never exceed 0.6. The internal lighting of the prim standard and all qualification indicators shall be subject to the approval of the qualifying activity.

3.14.1 Lamps. Lamps shall be in accordance with MIL-L-25467 and MIL-L-27160.

3.14.2 Prime standard. The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an attitude director conforming to the requirements of this specification or lighted model. The prime standard shall be equipped with both red and white lighting and shall be used as the reference for acceptance of all production indicators."

PAGE 26

* 3.14.2.1 Line 9 delete "The data and,". Line 9 change "if" to data shall be submitted in MIL-STD-831 format".

* 3.15.1 Line 2 delete "MIL-B-83292" and add: "MIL-B-81793".

PAGE 27

3.18.2 Delete and add:
"3.18.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, Type I or II as applicable, and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 30

Delete "4.4.1.1, a and b".

Delete "4.4.2, a, b, and c"

PAGE 31

Add:
"4.4.4 Test equipment compatibility tests. One test sample shall be shipped to the contracting activity for verification of test equipment compatability requirements specified in paragraphs 3.5.3.1 and 3.5.3.2. Fulfillment of this requirement shall be determined by complete successful verification of all test performed on the test sample, as specified for final acceptance in T.O.s 5F8-3-38-3 and 5F8-3-36-2 at depot and field levels respectively."

PAGE 40

4.6.12 Line 3: delete "1.0" and add "5.0".

PAGE 41

4.6.13 Delete and add:



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"4.6.13 Attitude warning flag. The indicator shall be properly connected, but no power applied. Power shall then be applied, and the flag shall disappear. The power shall be disconnected, and the flag shall appear. Power shall be applied, and the jumper from pin J to pin A shall be disconnected. The flag shall appear."

PAGE 43

4.6.22.1.1 Delete and add:
"4.6.22.1.1 Light. The indicator shall be subjected to the light tests (brightness, color, and stray light) as specified in MIL-L-27160 and MIL-L-25467".

PAGE 44

4.6.22.2.2 Delete and add:
"4.6.22.2.2 Brightness. With 2.50 ± 0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the prime standard, like areas of the pointers, command bars, alarm flags, sphere, numerals, indicia, markings, et cetera, of the indicator under test and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.5V shall fall within the brightness settings of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar in any individual area or the lighting of the indicator and prime standard is not uniform, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 and MIL-L-27160, as applicable. If the production units fail these tests, they shall be rejected and returned for rework. Qualifications units shall not be tested by use of the prime standard."

PAGE 48

4.6.35 Delete and add:
"4.6.35 Lighting circuit test. The indicator shall be subjected to all the tests as specified in MIL-L-25467 and MIL-I-27160, except the Life test shall be performed in conjunction with the Reliability tests."

PAGE 54

* Delete Section 5 in its entirety and add:

"5 PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.



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5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicators shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

PAGE 56

* Delete "6.4".

* 6.5 Line 11 and 12 delete "HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016" and add "OC ALC/MMEDO Tinker AFB OK 73145-5990".

* Delete "6.6".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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